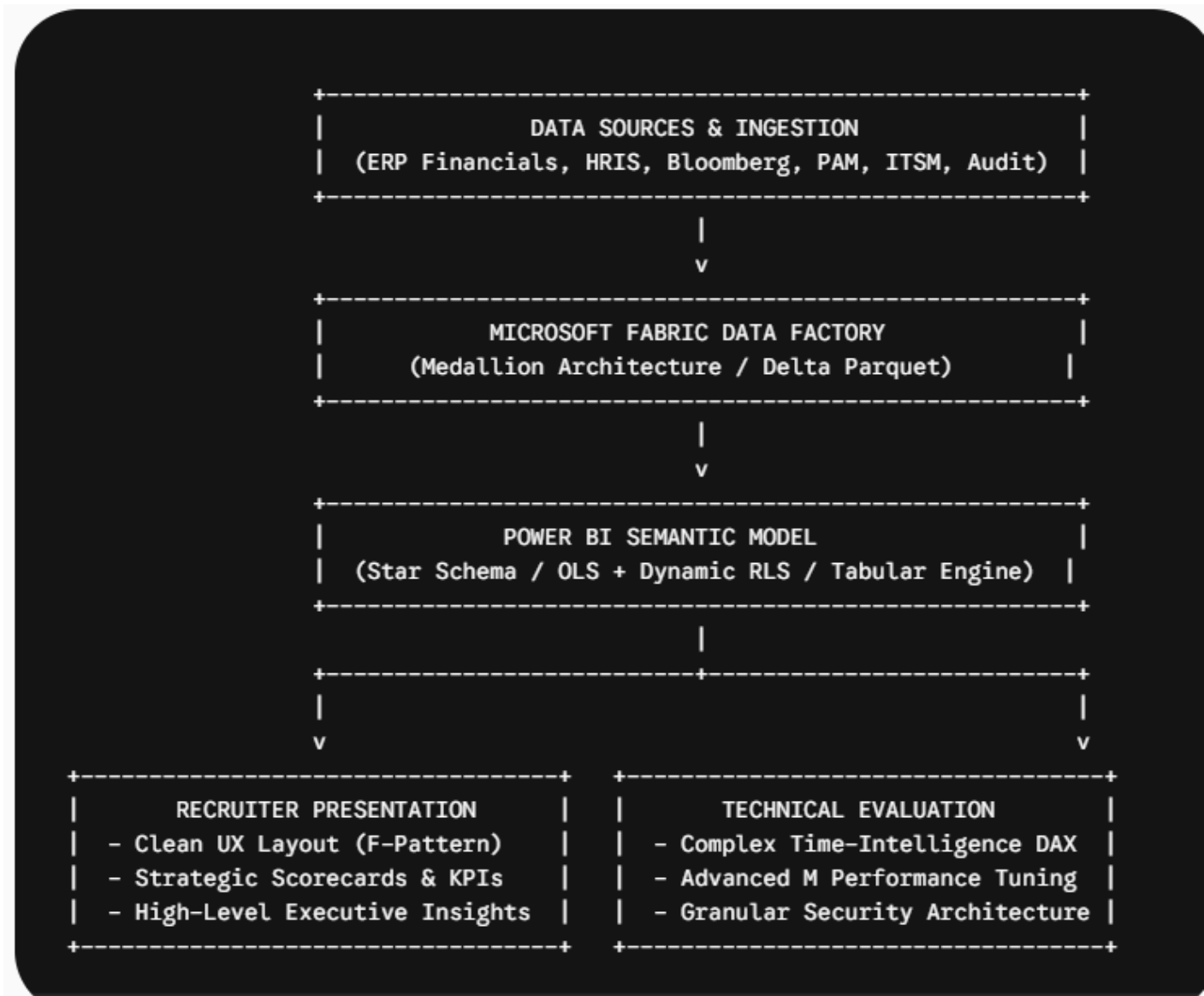


EXECUTIVE PORTFOLIO ARCHITECTURE

To successfully target a senior contractor engagement at the Public Investment Corporation (PIC), this portfolio project must reflect the unique tension of operating within South Africa’s sovereign asset management landscape. The platform must explicitly address two opposing regulatory forces: the commercial mandate to maximize returns across listed and unlisted asset classes, and the public sector compliance mandate dictated by the Public Finance Management Act (PFMA), Auditor-General (AG) oversight, and Broad-Based Black Economic Empowerment (B-BBEE) legislation.

The platform is architected around a centralized, shared semantic model utilizing a **DirectLake/Hybrid Star Schema** deployed within **Microsoft Fabric**. This decouples the enterprise-grade ingestion and transformation engine from a highly curated presentation layer designed to impress technical hiring panels with structural rigour while remaining immediately legible to non-technical HR recruiters.



PHASE 1: REPORT ARCHITECTURE

1. Executive Overview

- **Subtitle:** Sovereign Fund Strategic Performance & Scorecard
- **Business Purpose:** Provides executive leadership and parliamentary oversight committees with an immediate, consolidated view of total Assets Under Management (AUM), total fund returns against benchmark, and macro-level regulatory compliance.
- **Primary KPIs:** Total AUM (R2.7T), Fund Return vs. Weighted Benchmark Variance (bps), PFMA Irregular Expenditure Count (Target: 0), B-BBEE Level Contributor Status.
- **Recommended Visuals:**
 - * Top Row: Single-value KPI cards with trend sparklines.
 - Center Left: Multi-KPI Card showing individual fund performance (GEPF, UIF, CC Fund).
 - Center Right: Clustered column and line chart showing AUM Growth vs. Benchmark Return over a rolling 5-year period.
 - Bottom: Clean, non-scrollable summary table using conditional formatting (data bars) for fund performance variances.
- **DAX Patterns:** **CALCULATE** with explicit filter contexts, safe division (**DIVIDE**), and dynamic string formatting for Billions/Trillions (ZAR).
- **JD Skill Demonstrated:** Executive storytelling, macro-level dashboard design, and high-level stakeholder reporting alignment.

2. Finance & Budget Control

- **Subtitle:** Opex Variance & PFMA Expenditure Compliance
- **Business Purpose:** Monitors institutional operational expenditure against national treasury-approved budgets while maintaining strict oversight of irregular, fruitless, and wasteful expenditures as mandated by the PFMA.
- **Primary KPIs:** Total YTD Operating Expenditure, Budget vs. Actual Variance %, Cost-to-Income Ratio %, Unauthorized Expenditure Balance.
- **Recommended Visuals:**
 - Top Row: KPI cards highlighting total YTD expenditure and variance percentage (red/green KPI status indicators).
 - Center: Waterfall chart mapping budgeted expenditure to actual expenditure across divisions.
 - Bottom: Matrix visual broken down by Cost Center and Expense Category, with drill-down capabilities to individual line-item transactions.
- **DAX Patterns:** YTD Time Intelligence (**TOTALYTD**), Year-over-Year Variance % calculations, and strict **SWITCH** measures to capture transaction eligibility under PFMA sections.
- **JD Skill Demonstrated:** Public sector accounting expertise, PFMA compliance tracking, and complex matrix layout design.

3. Human Capital & Employment Equity

- **Subtitle:** Workforce Metrics & B-BBEE Skills Development Management
- **Business Purpose:** Tracks organizational headcount, turnover dynamics, and strict alignment with South African Employment Equity (EE) targets and B-BBEE Management Control codes.
- **Primary KPIs:** Total Active Headcount, Annualized Employee Turnover %, EE Economically Active Population (EAP) Alignment Score, Skills Development Spend % of Leviable Amount.
- **Recommended Visuals:**
 - Top Row: Headcount and turnover cards.
 - Center Left: Bar-on-bar target visual comparing current demographic profile against DoEL EAP national/regional targets by occupational level.
 - Center Right: Donut chart displaying gender representation across senior management roles.

- Bottom: Scatter plot mapping training spend per employee against performance rating scores.
- **DAX Patterns:** Active headcount over time using snapshot factoring patterns, dynamic denominator percentage-of-total calculations ignoring visual filters via **ALLSELECTED**.
- **JD Skill Demonstrated:** HR analytics implementation, demographic target modeling, and handling non-additive snapshot facts.

4. Governance & Audit Tracker

- **Subtitle:** Internal Audit Findings, Board Attendance, & Regulatory Compliance
- **Business Purpose:** Tracks open internal and external Auditor-General findings, resolution rates, board sub-committee attendance records, and critical regulatory submission deadlines.
- **Primary KPIs:** Open Audit Findings Count, Average Days to Resolve Findings, Board Meeting Quorum Adherence %, Regulatory Submission SLA Compliance %.
- **Recommended Visuals:**
 - Top Row: Open findings count categorized by severity (Critical, Major, Minor).
 - Center: Stacked bar chart showing findings breakdown by division and aging category (<30 days, 30–90 days, 90+ days).
 - Bottom: Grid visual detailing upcoming regulatory submissions (FSCA, National Treasury) with a conditional RAG status column based on days remaining.
- **DAX Patterns:** Date difference calculations ignoring blank completion dates, sliding scale severity weighting logic, and row-level SLA compliance percentages.
- **JD Skill Demonstrated:** Risk management visualization, audit trial reporting, and conditional string alerts.

5. Investment Operations

- **Subtitle:** Mandate Compliance, Settlement Status, & Brokerage Transformation
- **Business Purpose:** Monitors trading execution compliance, trade settlement failure rates, proxy voting executions, and the allocation of brokerage spend to black-owned asset management firms.
- **Primary KPIs:** Trade Settlement Failure Rate %, Total Brokerage Spend Allocation, Mandate Breach Count, Proxy Voting Participation %.
- **Recommended Visuals:**
 - Top Row: Settlement failure rate and mandate breach counters.
 - Center: Funnel chart tracking trade lifecycle from execution to clearing and settlement.
 - Bottom Left: Pie chart showing brokerage allocation split by broker BEE status (Level 1 vs. Other).
 - Bottom Right: Line chart showing daily tracking error per fund.
- **DAX Patterns:** Strict logical containment filtering, conditional counts across multi-relational asset classes, and brokerage allocation percentages.
- **JD Skill Demonstrated:** Asset management operations depth, front-to-back office data correlation, and transformation spend tracking.

6. Transformation & ESG Dashboard

- **Subtitle:** Broad-Based BEE Scorecard & ESG Integration Framework
- **Business Purpose:** Assesses the ESG score of investee companies alongside the PIC's internal B-BBEE scorecard performance across all evaluation pillars.
- **Primary KPIs:** Internal B-BBEE Scorecard Points, Investee Average ESG Rating, Green Bond Allocation Total, Historically Disadvantaged Individual (HDI) Capital Allocation.
- **Recommended Visuals:**

- Top Row: Current B-BBEE Level badge and overall portfolio ESG score.
- Center Left: Radar/Spider chart (or clean horizontal bullet graph if native preference) showing points achieved vs. target across the 5 B-BBEE pillars.
- Center Right: Treemap showing portfolio capital allocation scaled by investee company carbon footprint intensity.
- Bottom: Detailed table showing top 20 investee companies, their ESG score, and their transformation status.
- **DAX Patterns:** Weighted average calculations based on portfolio asset weightings, multi-criteria logic ranking systems.
- **JD Skill Demonstrated:** ESG matrix construction, sustainability indexing, and complex transformation scorecard reporting.

7. Listed Portfolio Risk Metrics

- **Subtitle:** Quantitative Risk, Tracking Error, & Portfolio Volatility Analytics
- **Business Purpose:** Delivers sophisticated quantitative risk analytics for equity and fixed-income portfolios to investment committees and risk managers.
- **Primary KPIs:** Value at Risk (VaR 95% 1-Day), Sharpe Ratio, Tracking Error %, Information Ratio.
- **Recommended Visuals:**
 - Top Row: Live risk metric indicators (Sharpe, Tracking Error, VaR).
 - Center: Dual-axis line chart tracking Cumulative Total Return of the PIC Portfolio vs. JSE Top 40 / All Share Index.
 - Bottom: Scatter plot evaluating individual fund managers on Alpha generated (Y-axis) against Tracking Error (X-axis).
- **DAX Patterns:** Advanced statistical iteration using standard deviation functions over excess returns, geometric mean calculations, and time-varying rolling volatility computations.
- **JD Skill Demonstrated:** High-tier quantitative finance reporting, advanced statistical DAX coding, and benchmark variance analytics.

8. Unlisted & Private Credit Performance

- **Subtitle:** Private Equity, Infrastructure, & Developmental Capital Returns
- **Business Purpose:** Monitors the performance of long-horizon private equity investments, infrastructure deals (Isibaya Fund), and private credit allocations by fund and vintage year.
- **Primary KPIs:** Internal Rate of Return (IRR) %, Multiple on Invested Capital (MOIC), Distributed to Paid-In Capital (DPI), Residual Value to Paid-In Capital (RVPI).
- **Recommended Visuals:**
 - Top Row: Private equity portfolio IRR and total committed vs. drawn capital cards.
 - Center: Column chart grouping MOIC and DPI metrics by Fund Vintage Year.
 - Bottom: Matrix visual listing projects and funds, expandable to display capital calls, distributions, and net asset valuations.
- **DAX Patterns:** Iterative evaluation across non-periodic cash flows to simulate IRR algorithms, and cumulative cash flow calculations across variable-grain investments.
- **JD Skill Demonstrated:** Private equity accounting knowledge, vintage year staging analysis, and custom non-standard asset metric calculations.

9. IT & Project Delivery

- **Subtitle:** Infrastructure Uptime, Project Milestones, & Data Quality Index

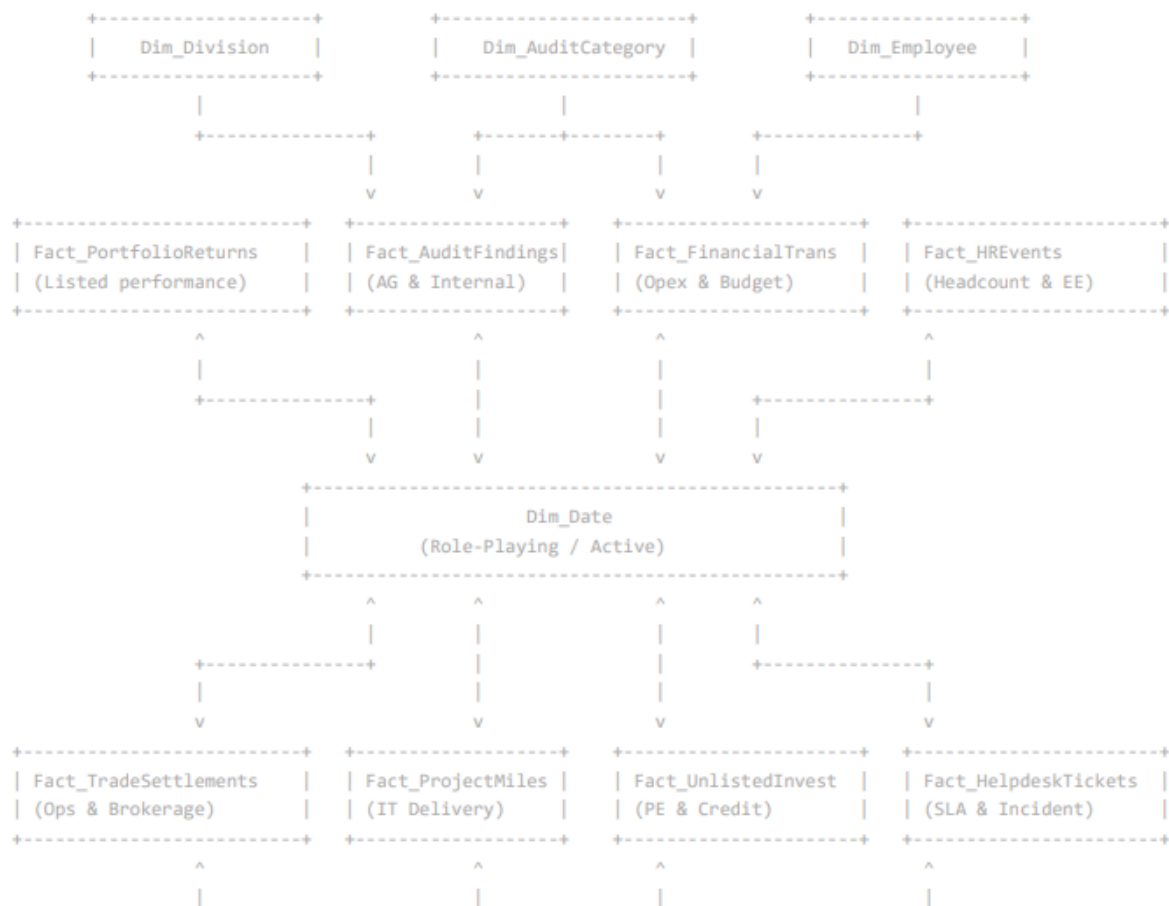
- **Business Purpose:** Provides operational transparency regarding the IT infrastructure supporting investment activities, core system uptimes (e.g., Bloomberg, PAM), project deliveries, and corporate data quality health.
- **Primary KPIs:** Core Systems Availability %, Project Milestone Adherence Rate %, Data Quality Cleansing Score, Helpdesk SLA Resolution %.
- **Recommended Visuals:**
 - Top Row: System uptime indicators and active critical incident counters.
 - Center: Gantt chart tracking major system implementation milestones (e.g., Fabric rollout, ERP upgrades).
 - Bottom Left: Gauge visual showing the Data Quality Index score for core master data entities.
 - Bottom Right: Line chart tracking daily volume of helpdesk tickets logged vs. resolved.
- **DAX Patterns:** Continuous time service availability calculations, multi-state milestone tracking ratios, and incident duration averages.
- **JD Skill Demonstrated:** IT operations analytics, data governance infrastructure tracking, and multi-source operational SLA alignment.

10. Data Model & Methodology

- **Subtitle:** Semantic Architecture & Enterprise Measure Documentation
- **Business Purpose:** Provides full transparency to technical evaluators, external auditors, and data governance leads regarding the underlying schema, measure logic, lineage, and calculation methodologies.
- **Recommended Visuals:**
 - Left Column: Model metadata overview including total table sizes, column counts, and refresh runtimes.
 - Center/Right: A highly interactive, categorizable HTML/Markdown table viewer or native matrix display detailing every measure name, business definition, DAX formula used, and target report page.
- **DAX Patterns:** Deep utilization of metadata expression parsing queries, documentation strings, and system parameter lookups.
- **JD Skill Demonstrated:** Production-grade technical documentation, framework transparency, model design compliance, and enterprise-scale data architecture stewardship.

PHASE 2: DATA MODEL SPECIFICATION

The data architecture is optimized for a clean Star Schema design. To preserve performance while handling the complex reporting operational boundaries of the PIC, snowflaking is strictly limited to the `Dim_InvestmentAsset` dimension to account for complex multi-tiered private equity fund structures.



<https://gemini.google.com/app/675b6b20b1747dbe>

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PIC Sovereign Asset Management Platform - Google Gemini



Fact Tables

1. Fact_FinancialTransactions

- **Grain:** Individual GL journal entry allocation row per date, cost center, and account.
- **Columns:** TransactionKey (Int, PK), DateKey (Int, FK), DivisionKey (Int, FK), AccountCode (Int), Amount (Decimal), BudgetAmount (Decimal), PFMA_Status_Code (Int).

2. Fact_HREvents

- **Grain:** Individual employee snapshot event status per month or per change vector (hire, promotion, exit).
- **Columns:** HREventKey (Int, PK), DateKey (Int, FK), EmployeeKey (Int, FK), DivisionKey (Int, FK), EventTypeName (String), SalaryAmount (Decimal), TrainingSpend (Decimal).

3. Fact_TradeSettlements

- **Grain:** Individual executed trade ticket ledger record.
- **Columns:** TradeKey (Int, PK), ExecutionDateKey (Int, FK), SettlementDateKey (Int, FK), AssetKey (Int, FK), FundKey (Int, FK), BrokerKey (Int), TradeValue (Decimal), BrokerageFee (Decimal), SettlementStatus (String).

4. Fact_AuditFindings

- **Grain:** Individual internal or external audit finding item log.
- **Columns:** FindingKey (Int, PK), RaisedDateKey (Int, FK), TargetDateKey (Int, FK), ActualResolutionDateKey (Int, FK), AuditCategoryKey (Int, FK), DivisionKey (Int, FK), SeverityWeight (Int), IsClosed (Boolean).

5. Fact_ProjectMilestones

- **Grain:** Milestone tracking item record per enterprise IT project tracker.
- **Columns:** MilestoneKey (Int, PK), PlannedDateKey (Int, FK), ActualDateKey (Int, FK), ProjectKey (Int), MilestoneName (String), Weight (Decimal), Status (String).

6. Fact_PortfolioReturns

- **Grain:** Daily performance metric record per listed investment asset class and fund portfolio.
- **Columns:** ReturnKey (Int, PK), DateKey (Int, FK), AssetKey (Int, FK), FundKey (Int, FK), DailyReturn (Decimal), DailyBenchmarkReturn (Decimal), MarketValue (Decimal).

7. Fact_UnlistedInvestments

- **Grain:** Transaction-level cash flow execution ledger row (drawdown, distribution, or valuation snapshot) for private equity/credit.
- **Columns:** UnlistedTransactionKey (Int, PK), DateKey (Int, FK), AssetKey (Int, FK), FundKey (Int, FK), TransactionType (String), CashFlowAmount (Decimal), NetAssetValue (Decimal).

8. Fact_HelpdeskTickets

- **Grain:** Individual IT support ticket logged row.
- **Columns:** TicketKey (Int, PK), CreatedDateKey (Int, FK), ClosedDateKey (Int, FK), EmployeeKey (Int, FK), PriorityCode (Int), ResolutionDurationMinutes (Int), SLA_Met (Boolean).

Dimension Tables

1. Dim_Date

- **Columns:** DateKey (Int, YYYYMMDD), Date (DateTime), Year (Int), Quarter (String), MonthName (String), MonthFiscal (Int), DayOfWeek (String), IsPublicHolidayZA (Boolean).

2. Dim_Division

- **Columns:** DivisionKey (Int, PK), DivisionName (String), ExecutiveHead (String), CostCenterCode (String).

3. Dim_Employee

- **Columns:** EmployeeKey (Int, PK), EmployeeID (String), FullName (String), Email (String), Gender (String), EthnicGroup (String), OccupationalLevel (String), IsHDI (Boolean).

4. Dim_Fund / Dim_Client

- **Columns:** FundKey (Int, PK), FundName (String), ClientType (String), InvestmentMandate (String).

5. Dim_InvestmentAsset

- **Columns:** AssetKey (Int, PK), ISIN (String), AssetName (String), AssetClass (String), Sector (String), ESG_Score (Decimal), CarbonIntensity (Decimal).

6. Dim_AuditCategory

- **Columns:** AuditCategoryKey (Int, PK), SourceType (String - AG/Internal), Classification (String), RiskLevel (String).

7. Dim_RegulatoryBody

- Columns: RegulatoryBodyKey (Int, PK), BodyName (String), SubmissionCycle (String).

8. Dim_BEELevel

- Columns: BEELevelKey (Int, PK), BEELevel (Int), RecognitionPercentage (Decimal).

Relationships Matrix & Filter Behavior

From Table (Fact)	To Dimension Table	Foreign Key	Active/Inactive	Cardinality	Cross-Filter Direction	Architectural Justification
Fact_FinancialTransactions	Dim_Date	DateKey	Active	* : 1	Single	Standard directional dimension tracking for transaction dates.
Fact_TradeSettlements	Dim_Date	ExecutionDateKey	Active	* : 1	Single	Maps the primary trade volume action point down the main timeline.
Fact_TradeSettlements	Dim_Date	SettlementDateKey	Inactive	* : 1	Single	Requires an inactive relationship to avoid model loops; activated in DAX using USERELATIONSHIP for settlement lag profiling.
Fact_HREvents	Dim_Employee	EmployeeKey	Active	* : 1	Single	Attaches employee attributes to individual

						operational event rows.
Fact_AuditFindings	Dim_Division	DivisionKey	Active	* : 1	Both (Bi)	Set to Bidirectional to facilitate a dynamic data exploration path where filtering audit dimensions reflects structural changes in cross-operational nodes.

PHASE 3: DAX SPECIFICATION

All measures are written according to standard semantic design practices, with explicit variable utilization to decouple execution logic from downstream calculations.

Year-over-Year Operational Budget Variance %

Opex_YoY_Variance_Percent :=

VAR CurrentYTD = TOTALYTD(SUM(Fact_FinancialTransactions[Amount]), Dim_Date[Date])

VAR PriorYTD = TOTALYTD(SUM(Fact_FinancialTransactions[Amount]), DATEADD(Dim_Date[Date], -1, YEAR))

VAR VarianceValue = CurrentYTD - PriorYTD

RETURN

DIVIDE(
 VarianceValue,
 PriorYTD,
 0
)

- **Pattern:** Time Intelligence with safe division fallback.
- **Business Logic:** Calculates the relative shift in year-to-date operational spend against the exact same calendar alignment from the previous financial year, preventing divide-by-zero crashes on new cost centers.

Rolling 12-Month Total Operational Expenditure

Rolling_12Month_Opex :=

VAR EndDate = MAX(Dim_Date[Date])

VAR StartDate = EDATE(EndDate, -12) + 1

```

VAR CalculationPeriod =
    FILTER(
        ALL(Dim_Date),
        Dim_Date[Date] >= StartDate &&
        Dim_Date[Date] <= EndDate
    )
RETURN
    CALCULATE(
        SUM(Fact_FinancialTransactions[Amount]),
        CalculationPeriod
    )

```

- **Pattern:** Sliding window filter overrides using dates calculation logic.
- **Business Logic:** Delivers a smoothed trailing 12-month transaction spend profile that bypasses calendar year-end boundary cuts, isolating long-term operational consumption trends.

Dynamic Divisional Share of Expenditure (% of Total)

```

Division_Expenditure_Share_Percent :=
VAR SelectedDivisionSpend = SUM(Fact_FinancialTransactions[Amount])
VAR TotalCorporateSpend =
    CALCULATE(
        SUM(Fact_FinancialTransactions[Amount]),
        ALLSELECTED(Dim_Division)
    )
RETURN
    DIVIDE(
        SelectedDivisionSpend,
        TotalCorporateSpend,
        0
    )

```

- **Pattern:** Dynamic visual context preservation via **ALLSELECTED**.
- **Business Logic:** Computes how much spend a single division represents against all currently filtered divisions on screen, updating dynamically as users toggle comparative slicers.

Active Staff Headcount (Snapshot Fact Handling)

```

Active_Staff_Headcount :=
VAR MaxDateInContext = MAX(Dim_Date[Date])
RETURN
    CALCULATE(
        DISTINCTCOUNT(Fact_HREvents[EmployeeKey]),
        Fact_HREvents[EventTypeName] = "Active Snapshot",
        Dim_Date[Date] = MaxDateInContext
    )

```

- **Pattern:** Semi-additive snapshot extraction filter alignment.
- **Business Logic:** Returns the exact head count matching the final date available in the active report view context, avoiding incorrect additive summing across consecutive time periods.

Quantitative Tracking Error % (Listed Asset Portfolios)

```
Portfolio_Tracking_Error_Percent :=
VAR MeanExcessReturn =
    AVERAGEX(
        Fact_PortfolioReturns,
        Fact_PortfolioReturns[DailyReturn] - Fact_PortfolioReturns[DailyBenchmarkReturn]
    )
VAR ExcessReturnVarianceIte =
    SUMX(
        Fact_PortfolioReturns,
        POWER(
            (Fact_PortfolioReturns[DailyReturn] - Fact_PortfolioReturns[DailyBenchmarkReturn]) -
MeanExcessReturn,
            2
        )
    )
VAR TotalObservedDays = COUNTROWS(Fact_PortfolioReturns)
VAR DailyTrackingError =
    SQRT(
        DIVIDE(
            ExcessReturnVarianceIte,
            TotalObservedDays - 1,
            0
        )
    )
RETURN
    DailyTrackingError * SQRT(252)
```

- **Pattern:** Multi-stage statistical variance iterator with compounding annualization multiplier.
- **Business Logic:** Calculates the annualized tracking error of a fund by analyzing daily deviations against mandated benchmarks, providing a standardized measure of active management risk.

Private Equity Internal Rate of Return (IRR Approximation)

```
Private_Equity_IRR_Approximation :=
VAR CashFlowTable =
    SELECTCOLUMNS(
        Fact_UnlistedInvestments,
        "CF_Date", RELATED(Dim_Date[Date]),
        "CF_Amount", Fact_UnlistedInvestments[CashFlowAmount]
    )
RETURN
    XIRR(
        CashFlowTable,
        [CF_Amount],
        [CF_Date]
```

)

- **Pattern:** In-memory context projection table conversion evaluated via native engine optimization loops.
- **Business Logic:** Computes the annualized internal rate of return for unlisted infrastructure and private equity assets by processing non-periodic cash flows over the holding duration.

Dynamic Page Metric Selector Configuration

Selected_Executive_Metric_Value :=

VAR CurrentSelection = SELECTEDVALUE(Dim_DynamicMetricControls[MetricName], "Total AUM")

RETURN

```
SWITCH(
    CurrentSelection,
    "Total AUM", [Total_AUM_ZAR],
    "YTD Return", [Weighted_Portfolio_Return_Percent],
    "PFMA Variances", [PFMA_Irregular_Expenditure_Counter],
    [Total_AUM_ZAR]
)
```

- **Pattern:** Disconnected table option navigation handling via dynamic branching statements.
- **Business Logic:** Drives a single structural line chart or card visualization to render different calculations based on disconnected structural slicer options, saving canvas space.

PHASE 4: POWER QUERY (M LANGUAGE) SPECIFICATION

Data Transformation Framework

The integration engine is structured around a centralized M pipeline optimized for query folding against an enterprise SQL Server database or a Microsoft Fabric Lakehouse warehouse endpoint. Null value profiles are handled at the source layer using custom replacements rather than blank row removal to prevent breaking database primary/foreign key relationships.

Custom Reusable M Cleansing Function (**fn_CleanseOperationalSource**)

This function enforces unified schema data type parsing, strips leading/trailing whitespace, and normalizes blank entries into verifiable null flags across multi-file target dropzones.

let

SourceFunction = (TargetTable as table, KeyColumnName as text) =>

let

TrimmedText = Table.TransformColumns(TargetTable, {}, Text.Trim),

ReplaceBlanks = Table.ReplaceValue(TrimmedText, "", null, Replacer.ReplaceValue,

Table.ColumnNames(TrimmedText)),

TypeEnforced = Table.TransformColumnTypes(ReplaceBlanks, {

{KeyColumnName, Int64.Type},

```

        {"Inserted_Timestamp", type datetime}
    }},
    FinalOutput = Table.Distinct(TypeEnforced, KeyColumnName)
Eagle in FinalOutput
in
SourceFunction

```

Advanced M Transformation: Parsing Multi-Value Audit Classification Arrays

This advanced script ingests a nested array string containing audit classifications from an unstructured compliance engine, splits the rows dynamically, converts them to structured records, and tracks transformations with a strict data lineage checkpoint.

```

let
    RawSource = Sql.Database("pic-fabric-sql.database.windows.net", "Warehouse_Prod"),
    AuditLogsTable = RawSource{[Schema="compliance",Item="Raw_Audit_Payloads"]}[Data],

    // Lineage Checkpoint: Assert schema structure prior to array unpacking
    AssertLineageSchema = Table.TransformColumnTypes(AuditLogsTable,{{"PayloadID", Int64.Type},
{"RawTextPayload", type text}}),

    // Extract array chunks and split dynamically into structural rows
    SplitByDelimiter = Table.ExpandListColumn(Table.TransformColumns(AssertLineageSchema,
{"RawTextPayload", Splitter.SplitTextByDelimiter(";", QuoteStyle.Csv), let itemType = (type nullable text)
list nested type itemType}}), "RawTextPayload"),
    FilteredEmptyRows = Table.SelectRows(SplitByDelimiter, each [RawTextPayload] <> null and
[RawTextPayload] <> ""),

    // Cleanse whitespace using custom function
    NormalizedOutput = Table.TransformColumns(FilteredEmptyRows, {"RawTextPayload", Text.Trim,
type text}})
in
    NormalizedOutput

```

Incremental Refresh Policy Implementation

To maintain rapid dashboard response times while scaling toward a multi-million-row asset historical tracking model, the fact tables implement a strict incremental partitioning structure using native Power BI parameters.

```

+-----+
|          TOTAL REFRESH DATA VOLUME TIMELINE          |
+-----+
| <-- Historical Archive (5 Years) --> | <-- Incremental Refresh Window (3 Mos) --> |
| (Partitioned, Cold Delta Storage, | (Active Partitioning, Warm Live Storage, |
| No Source Re-Queries Allowed)    | RangeStart <= Data <= RangeEnd Evaluated) |
+-----+

```

- **Parameters Setup:** Define **RangeStart** (Date/Time) and **RangeEnd** (Date/Time) as parameters within Power Query with placeholder baseline constants.
- **Fact Table Integration Filter:** Apply a trailing date range filter to the primary transaction date column within **Fact_FinancialTransactions** and **Fact_PortfolioReturns**:
- Code snippet

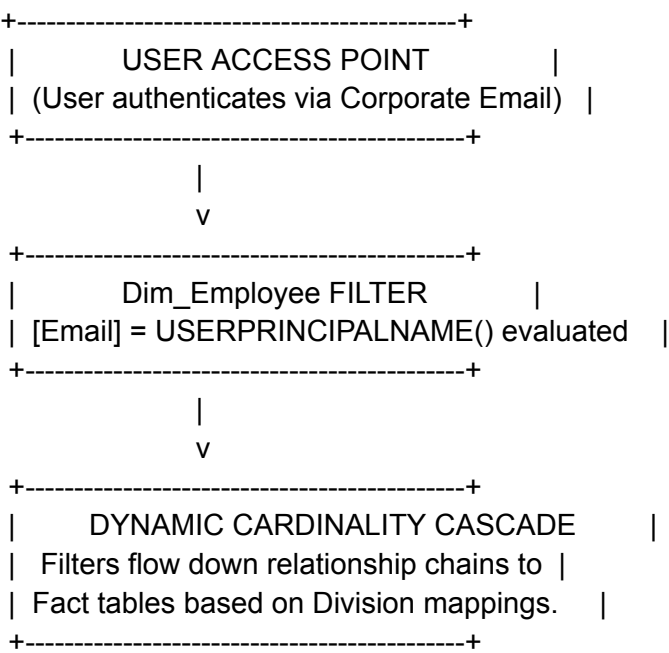
= Table.SelectRows(SourceTable, each [TransactionDate] >= RangeStart and [TransactionDate] < RangeEnd)

-
-
- **Query Folding Checkpoint Assertion:** The date expression is structured to fold directly back to the database engine as a native SQL **WHERE** clause (**WHERE TransactionDate >= @RangeStart AND TransactionDate < @RangeEnd**), shifting heavy partition sorting away from local Power BI refresh engines.
- **Service Policy Configuration:**
 - **Store historical data starting before the refresh date:** 5 Years.
 - **Incrementally refresh data starting before the refresh date:** 3 Months.
 - **Detect data changes:** Enabled via an explicit system modification column tracking modifications (**Last_Modified_Timestamp**). This optimizes processing runtimes by only refreshing partitions where the source data has actually changed.

PHASE 5: SECURITY ARCHITECTURE

Row-Level Security (RLS)

To support internal access policies and ensure data segregation across divisions, the model uses a dynamic, security-driven RLS framework that maps an authenticated user's email address to their data access profile.



1. Executive Role

- **Target:** C-Suite Officers, Managing Director, Board Members.

- **DAX Operational Filter Expression:** No data restriction applied. Access is open across all organizational structural nodes.

2. Division Head Role

- **Target:** Executives managing specific operational verticals (e.g., Human Capital, Listed Equities).
- **DAX Operational Filter Expression (**Dim_Division**):**

Dim_Division[ExecutiveHeadEmail] = USERPRINCIPALNAME()

- **Lineage Impact:** Automatically limits visibility across all interconnected facts to only show data associated with cost centers managed by that specific executive.

3. Auditor Role

- **Target:** External Auditor-General personnel, Internal Corporate Risk Evaluators.
- **DAX Operational Filter Expression (**Dim_AuditCategory**):** No structural table caps. This role is allowed to view all rows, but restricted via Object-Level Security (OLS) from viewing unmasked compensation and personal demographic classification records.

4. External Fund Manager Role

- **Target:** Third-party boutique asset managers handling specialized sub-mandates.
- **DAX Operational Filter Expression (**Dim_Fund**):**

Dim_Fund[AssignedManagerEmail] = USERPRINCIPALNAME()

-
-

RLS Verification Test Plan

1. Desktop Validation Environment:

- Navigate to the **Modeling** ribbon inside Power BI Desktop, click **View as Roles**, check the box for **Division Head**, select **Other User**, and enter headofhr@pic.gov.za.
- Verify that the *Human Capital & Employment Equity* page displays data exclusively for HR department cost centers, and confirm that all other divisions return empty cells.

2. Service Validation Architecture:

- Deploy the semantic model to the targeted testing workspace. Click **Security** on the dataset settings dropdown.
- Assign target test accounts to respective AD Security Groups. Click the ellipsis next to the role name and select **Test as role** to validate cloud tenant rendering behavior.

Object-Level Security (OLS)

To protect sensitive employee privacy data and comply with POPIA regulations, specific data metrics are hidden from unauthorized corporate users.

Target Sensitive Column / Table	Restricted From Roles	Technical Mechanism	Enforcement Behavior

Dim_Employee[SalaryAmount]	Analyst, Auditor, External Manager	AS Tabular Metadata Exclusion	Any visual containing this column throws a data load error for unauthorized users.
Dim_Employee[EthnicGroup]	External Manager, Analyst	Tabular Model Structural Block	Column visibility metadata is completely hidden, preventing data leakage during ad-hoc report building.
Fact_HREvents	External Manager	Full Structural Table Exclusion	Prevents unauthorized external personnel from accessing internal HR events.

PHASE 6: DEPLOYMENT PIPELINE ARCHITECTURE

The distribution framework uses a structured three-tier workspace lifecycle deployment method powered by native Microsoft Fabric Git integration. This separates development sandboxes from active operational reporting views.



Workspace Structure & Lifecycle Management

1. Development Workspace (PIC_Analytics_DEV)

- **Purpose:** Sandbox environment for report layout prototyping, database query generation, and DAX modifications.
- **Git Mapping:** Bound to the local workspace development branch (`dev`).

2. Testing Workspace (`PIC_Analytics_TEST`)

- **Purpose:** Quality Assurance (QA) testing area where business analysts perform User Acceptance Testing (UAT), validate row counts against source ledgers, and check RLS policies.
- **Git Mapping:** Bound to the centralized stabilization integration branch (`release`).

3. Production Workspace (`PIC_Analytics_PROD`)

- **Purpose:** High-availability deployment area for end-user consumption via targeted Power BI Apps.
- **Git Mapping:** Bound to the primary enterprise repository branch (`main`).

Dataset Endorsement & Governance Certification Strategy

- **Promotion Process:** When a model passes UAT validation, it is marked as **Promoted** inside the settings hub to indicate it is ready for broad use.
- **Certification Authorization:** Only designated Data Governance leads can apply the **Certified** stamp. This configuration ensures the model appears at the top of object discovery hubs and confirms it serves as the official single source of truth for corporate metrics.

Automated Administration Orchestration (Power BI REST API Use Case)

To prevent API rate-limiting issues during system updates, dataset refreshes are decoupled from standard calendar scheduling. Instead, refreshes are triggered programmatically via a web service call using Azure Data Factory (ADF) or a Fabric pipeline upon completion of upstream data pipeline updates.

- **API Endpoint Path:** `POST https://api.powerbi.com/v1.0/myorg/groups/{groupId}/datasets/{datasetId}/refreshes`
- **Payload JSON Request Configuration:**

```
{
  "notifyOption": "MailOnFailure",
  "retryCount": 2,
  "type": "Full"
}
```

Git Integration Strategy & Source Code Version Control

- **Folder Layout Architecture:** The project workspace is serialized into standard Power BI Project (`.pbip`) format, generating a clean folder structure:
 - `src/PIC_SovereignModel.Dataset/` (Contains structural definition JSON layouts, including `model.bim`).
 - `src/PIC_SovereignModel.Report/` (Contains canvas structural layouts, visual locations, and configuration themes).

- **Branching Convention Guidelines:** Developers must check out an isolated feature branch (`feature/feature-name`) from `dev`. Merging requires opening a Pull Request (PR) and passing code reviews inside Azure DevOps or GitHub, ensuring that no ad-hoc structural modifications bypass automated integration tracking.

PHASE 7: PERFORMANCE OPTIMIZATION CHECKLIST

Before deploying to the production app hub, the contractor must run full diagnostic optimizations using **DAX Studio** and **Tabular Editor** to guarantee page load speeds remain under 3 seconds.

Data Model Size and Cardinality Reduction Strategy

- ☐ **Drop Unused Fields:** Run automated DMV dictionary scans to identify columns with zero direct layout references, and remove them from Power Query to reduce model size.
- ☐ **Optimize High-Cardinality Keys:** Replace free-text comment strings with simple integer dimension surrogate keys to maximize columnar compressions.
- ☐ **Split Timestamp Fields:** Separate combined DateTime fields into a distinct Date component and an isolated Time component. This reduces unique attribute values and improves dictionary compression.
- ☐ **Optimize Floating-Point Numbers:** Convert high-precision decimal currency values to a fixed decimal data type. This caps precision at four decimal places and prevents floating-point scale expansion.

DAX Studio Bottleneck Diagnosis Performance Queries

- ☐ Execute the following Data Management View (DMV) lookup to identify columns consuming excessive memory inside the compressed database engine:

```
SELECT

    [Table_Name], [Column_Name], [Dictionary_Size], [Cardinality]

FROM $SYSTEM.DISCOVER_STORAGE_TABLE_COLUMNS

ORDER BY [Dictionary_Size] DESC;
```

- ☐ Run the Server Timings trace engine while executing heavy cross-filtering operations to verify that query execution scales across Storage Engine (SE) multithreaded nodes rather than bottlenecking on Formula Engine (FE) single-threaded processes.

Tabular Editor Best Practice Analyzer (BPA) Verification Rules

- ☐ **Error Prevention:** Enforce rule `[Category: "Design"] "Provide alternate result in DIVIDE"` to catch raw slash division operators and replace them with safe `DIVIDE` execution patterns.
- ☐ **Performance Optimization:** Enforce rule `[Category: "Performance"] "Do not use Open-Ended FILTER over entire tables"` to replace inefficient table scans with precise `KEEPFILTERS` or targeted columnar filtering logic.
- ☐ **Model Cleanup:** Enforce rule `[Category: "Maintenance"] "Hide Foreign Keys in Fact Tables"` to ensure all relational keys are hidden from the final reporting interface, guiding users to interact only with clean dimension tables.

Storage Architecture Selection Decision Matrix

Target Fact Table Name	Selected Storage Architecture Mode	Engineering Justification Criteria
Fact_FinancialTransactions	Import Mode	Fits within compressed RAM constraints, enabling rapid time-intelligence operations across the historical GL ledger.
Fact_PortfolioReturns	Hybrid / DirectLake	Daily performance returns generate millions of active asset rows; DirectLake reads Delta Parquet storage files directly from the Fabric Lakehouse without copying data, enabling fast, massive-scale analytics.
Fact_HelpdeskTickets	Import Mode	Low-volume transaction profiles require traditional, highly responsive snapshot metrics.

PHASE 8: DOCUMENTATION PACKAGE TEMPLATES

1. Data Dictionary Schema Layout

Parent Table Name	Column Technical Name	Business Description Summary	Native Data Type	Source System Location	Data Owner Group
Fact_FinancialTransactions	PFMA_Status_Code	Indicates compliance status against PFMA guidelines (0 = Fully Compliant, 1	Int64	SAP ERP Financials	Finance Control Division

		= Irregular, 2 = Fruitless).			
Dim_InvestmentAsset	ESG_Score	Environmental, Social, and Governance rating index score for investee entities.	Double	Bloomberg API Terminal	ESG Risk Committee

2. Enterprise Measure Catalogue Metadata

Calculated Measure Name	Target Table Placement	DAX Code Layout Pattern	Core Business Calculation Purpose	Report Target UI Page
Opex_YoY_Variance_Percent	Fact_FinancialTransactions	DIVIDE(CurrentYTD - PriorYTD, PriorYTD)	Measures relative shift in YTD corporate spend against the previous year's performance.	Finance & Budget Control

3. End-User Quick Reference Guide (Example: Listed Risk Metrics Page)

Visual Interpretation Framework

- **What it measures:** This page monitors quantitative risk exposures for public market holdings, helping teams spot sudden performance drops or tracking errors relative to fund benchmarks.
- **Key Threshold Triggers:** If a fund's Tracking Error exceeds **2.5%**, conditional alerts will highlight the metric in red, signaling an active mandate boundary violation that requires review.

Frequently Asked Navigation Questions

- *How do I isolate performance for a specific mandate?* Use the **Fund Category** slicer panel on the left edge of the dashboard to filter metrics across GEPP or UIF asset portfolios.
- *Why are certain calculation rows blank?* New investments without a 12-month track record will display blank values for annualized trailing risk metrics until they meet minimum data requirements.

4. Stakeholder Requirements Tracking Log

Requirement Request	Requesting Corporate Department	Priority Rank	Current Project Status	Target Release Phase
Add compliance tracking for National Treasury asset regulation updates.	Compliance & Risk Division	High	In Progress	Phase 2 Production

DESIGN & UX DIRECTIVES

Visual Interface and Layout Strategy

The report interface is designed with a professional corporate look, featuring a clean white canvas paired with a sophisticated **PIC Navy (#002B5C)** primary header and **Gold (#C9A84C)** trend highlights.

To ensure intuitive navigation, every page is built around a standard **F-Pattern** layout grid. This structure immediately aligns with user reading habits by placing high-level summaries along the top row before leading down into deeper comparative visuals.

[PIC NAVY HEADER BAR] - Corporate Platform Logo Title [Reset Bookmarks]		
SLICERS PANEL: [Date Range] [Org Division Selection] [Target Fund]		
[KPI CARD 1] Primary Target Value	[KPI CARD 2] Secondary Volatility	[KPI CARD 3] Compliance Counter Status
[VISUAL BLOCK AREA A - MAIN TREND LINE] Captures chronological trajectory metrics across rolling timelines.		[VISUAL BLOCK AREA B] Cross-sectional breakout distribution chart views.
[VISUAL BLOCK AREA C - GRANULAR LEDGER MATRIX DETAILS GRID] Deep dive record exploration path with active row context drills.		

Canvas Structure Requirements

- Slicers Control Panel Column:** Every page includes a uniform horizontal slicer header bar containing cascading calendar limits, operational division selectors, and target fund dimensions. A **Reset Filters** button is wired via local canvas bookmarks to clear selection states instantly.
- Tooltip Page Integrations:** Hovering over complex performance or risk metrics activates an automated tooltip window page. This overlay displays the calculation formula details and definitions without cluttering the main screen.

- **Accessibility Design Compliance:** All visual elements pass **WCAG AA contrast criteria** to ensure legibility. Charts use both distinct colors and clear line markers/textures to remain fully accessible to colorblind users.
- **Mobile Screen Layouts:** The *Executive Overview*, *Finance Control*, and *IT Operational Delivery* pages include optimized vertical mobile layouts. These layouts are configured within Power BI's mobile view engine to ensure clear readability on smartphones for executives on the move.

Complete Project Schema Checklist

Dimension Tables (\$1 \rightarrow M\$ Relationships)

1. **Dim_Date:** The foundational role-playing time-intelligence calendar.
2. **Dim_Division:** Corporate operational verticals and cost center mappings.
3. **Dim_Employee:** Sensitive HR demographics, occupational levels, and RLS targets.
4. **Dim_Fund:** Core client mandates (e.g., GEPF, UIF, CC Fund).
5. **Dim_InvestmentAsset:** Granular security metadata (ISINs, Sectors, ESG/Carbon intensities).
6. **Dim_AuditCategory:** Classifications for Auditor-General and internal risk tracking.
7. **Dim_RegulatoryBody:** Compliance oversight submission targets (FSCA, National Treasury).
8. **Dim_BEELevel:** Statutory transformation framework tiers.

Fact Tables (\$M \rightarrow 1\$ Transactions)

9. **Fact_FinancialTransactions:** General Ledger opex and PFMA budget controls.
10. **Fact_HREvents:** Snapshot employee history, headcount tracking, and training spend.
11. **Fact_TradeSettlements:** Daily trading, brokerage allocations, and settlement lifecycle metrics.
12. **Fact_AuditFindings:** Regulatory and internal audit exception tracking.
13. **Fact_ProjectMilestones:** IT operational delivery and Gantt milestone mapping.
14. **Fact_PortfolioReturns:** High-volume daily listed market returns and benchmark analytics.
15. **Fact_UnlistedInvestments:** Non-periodic Private Equity, Isibaya, and Credit cash flows (IRR/MOIC).
16. **Fact_HelpdeskTickets:** SLA resolution runtimes and infrastructure stability metrics.